

Preclass Learning: APIs P2 - Authentication and Rate Limits (ICA 14)

1. Authentication and Authorization

- **API Key:** A secret token used to identify the client making the request, often passed as a query parameter or in the request header. Essential for tracking usage and preventing unauthorized access.
- **OAuth 2.0:** A more complex, secure standard where the client obtains an access token from an authorization server, allowing access to protected resources without exposing user credentials.

2. Handling Responses

The server's response is crucial for debugging and error handling.

- **HTTP Status Codes:** Standardized codes indicating the result of the request:
 - 200 OK: Request succeeded.
 - 400 Bad Request: Client error (e.g., missing parameter).
 - 401 Unauthorized: Authentication failed (bad API key).
 - 404 Not Found: The resource does not exist.
 - 500 Internal Server Error: Server-side problem.
- **Parsing Nested Data:** JSON structures can be deeply nested. Tools must be used to flatten or extract data from specific nested layers to create a tidy data frame.

3. Rate Limiting and Compliance

- **Rate Limits:** The maximum number of requests a client can make within a specific time window (e.g., 60 requests per minute).
- **Respecting Limits:** It is mandatory to read and adhere to the API's rate limit documentation, often by using automated delays (`System.Threading.Tasks.Task.Delay()`) between requests to avoid being blocked.
- **Terms of Service:** Reviewing the API's terms to ensure compliance regarding data usage, storage, and attribution.